

DG INTERNS HUB
CYBERSECURITY INTERNSHIP PROGRAM

WEEK 1 ASSIGNMENT REPORT

Networking Fundamentals

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PART 1

Objective

The goal of Week 1 of the DG Interns Hub Cybersecurity Internship is to build a strong foundation in computer networking and to understand how real-world systems communicate with each other. Since networking is the backbone of every cybersecurity discipline — from penetration testing to incident response — this week focuses on the theoretical models that describe network communication, the protocols that make the internet function, and hands-on practice using real network commands and Cisco Packet Tracer.

This report documents the theoretical research and practical exercises completed during Week 1, including screenshots of live command output and a simulated network built in Cisco Packet Tracer.

PART 2

The OSI Model

The Open Systems Interconnection (OSI) Model is a conceptual framework that standardizes how different networking systems communicate, dividing the process into seven distinct layers. Each layer has a specific function and hands off data to the layer above or below it.

Layer	Function	Real-Life Example
7. Application	Provides network services directly to end-user applications.	Web browsers, email clients (e.g., loading a webpage)
6. Presentation	Translates, encrypts, and compresses data for the application layer.	SSL/TLS encryption, JPEG/PNG image formatting
5. Session	Establishes, manages, and terminates connections (sessions) between devices.	A video call staying connected during a Zoom meeting
4. Transport	Ensures reliable, end-to-end delivery of data; handles segmentation.	TCP ensuring a file download arrives complete and in order
3. Network	Handles logical addressing and routing of packets across networks.	A router deciding the best path for data between cities
2. Data Link	Handles physical addressing (MAC) and error detection on the local link.	A switch forwarding frames to the correct device on a LAN
1. Physical	Transmits raw bits over a physical medium (cables, radio waves).	Ethernet cables, Wi-Fi signals, fiber optics

Understanding the OSI Model helps in troubleshooting network issues by isolating the layer where a problem occurs — for example, a cable fault is a Physical Layer issue, while a website not loading despite an active connection could be an Application Layer issue.

The TCP/IP Model

The TCP/IP Model is the practical, four-layer model that the modern internet actually runs on. It condenses the seven OSI layers into four, but achieves the same overall purpose: enabling reliable communication between devices.

TCP/IP Layer	OSI Equivalent	Examples
4. Application	Combines OSI's Application, Presentation, and Session layers	HTTP, HTTPS, FTP, DNS, SMTP
3. Transport	Same as OSI Transport Layer	TCP, UDP
2. Internet	Equivalent to OSI Network Layer	IP, ICMP, ARP
1. Network Access	Combines OSI's Data Link and Physical layers	Ethernet, Wi-Fi (802.11)

Comparison Summary:

- The OSI Model is a theoretical, 7-layer reference model used mainly for teaching and troubleshooting.
- The TCP/IP Model is a practical, 4-layer model that is actually implemented in real networks and the internet.
- Every protocol used in TCP/IP (like HTTP, TCP, IP) maps onto one or more layers of the OSI Model.

Core Networking Protocols

Task 3 required studying four essential protocols that power everyday internet communication.

HTTP / HTTPS

What it does: HTTP (HyperText Transfer Protocol) governs how web browsers and servers exchange data such as webpages. HTTPS is the secure version, encrypting the data using SSL/TLS so it cannot be intercepted or tampered with.

Where it is used: Every time a website is visited (e.g., <https://www.google.com>), the browser uses HTTPS to securely request and receive the page content.

FTP (File Transfer Protocol)

What it does: FTP is used to transfer files between a client and a server over a network, supporting uploading, downloading, and directory management.

Where it is used: Website administrators use FTP (or its secure version SFTP) to upload files to a web server, and organizations use it to share large files internally.

DNS (Domain Name System)

What it does: DNS translates human-readable domain names (like [google.com](https://www.google.com)) into machine-readable IP addresses (like 142.250.190.14), acting as the 'phonebook of the internet.'

Where it is used: Used every time a URL is typed into a browser — the DNS system resolves it to the correct server IP address before the connection is made.

DHCP (Dynamic Host Configuration Protocol)

What it does: DHCP automatically assigns IP addresses, subnet masks, gateways, and DNS servers to devices joining a network, removing the need for manual configuration.

Where it is used: Home routers and office networks use DHCP so that every laptop, phone, or IoT device connecting to Wi-Fi automatically receives a valid, unique IP address.

IP Addressing

What is an IP Address?

An IP (Internet Protocol) address is a unique numerical label assigned to every device connected to a network. It allows devices to locate and communicate with each other, similar to how a postal address identifies a house. The most common formats are IPv4 (e.g., 192.168.1.10) and IPv6 (a longer, hexadecimal format designed to handle the growing number of internet-connected devices).

Public vs. Private IP Addresses

Type	Description	Example Range
Private IP	Used within local networks (homes, offices); not routable on the internet.	192.168.0.0 – 192.168.255.255
Public IP	Assigned by an ISP; unique across the entire internet and used to communicate externally.	Assigned dynamically by ISP

Basic Subnetting Concept

Subnetting is the process of dividing a large network into smaller, more manageable sub-networks (subnets). This is done using a subnet mask (e.g., 255.255.255.0), which separates the IP address into a network portion and a host portion.

- It improves network performance by reducing broadcast traffic.
- It enhances security by isolating groups of devices from one another.
- Example: 192.168.1.0/24 with subnet mask 255.255.255.0 allows for 254 usable host addresses on that subnet.

PART 3

Network Commands

The following commands were executed in the terminal/command prompt to observe real network behavior, and the output was captured via screenshot.

1. ping google.com

Purpose: Tests connectivity to a remote host and measures round-trip time by sending ICMP echo requests.

SCREENSHOT

Terminal output of 'ping google.com' showing reply times and packet statistics

```
(mahnoor@kali)-[~/Desktop/dg]
$ ping google.com
PING google.com (142.250.203.174) 56(84) bytes of data.
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=1 ttl=128 time=49.8 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=2 ttl=128 time=52.1 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=3 ttl=128 time=52.3 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=4 ttl=128 time=48.9 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=5 ttl=128 time=48.7 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=6 ttl=128 time=49.4 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=7 ttl=128 time=51.7 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=8 ttl=128 time=53.0 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=9 ttl=128 time=49.2 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=10 ttl=128 time=56.4 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=11 ttl=128 time=52.1 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=12 ttl=128 time=54.1 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=13 ttl=128 time=51.6 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=14 ttl=128 time=51.6 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=15 ttl=128 time=140 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=16 ttl=128 time=156 ms
```

2. ipconfig / ifconfig

Purpose: Displays the current network configuration of the device — including IP address, subnet mask, and default gateway (ipconfig on Windows, ifconfig on Linux/macOS).

SCREENSHOT

Terminal output of 'ipconfig' or 'ifconfig' showing IP address, subnet mask, and gateway

```
(mahnoor@kali)-[~/Desktop/dg]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether 00:0c:29:58:3f:ec txqueuelen 1000 (Ethernet)
    RX packets 9 bytes 2796 (2.7 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.61.129 netmask 255.255.255.0 broadcast 192.168.61.255
    inet6 fe80::20c:29ff:fe58:3ff6 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:58:3f:f6 txqueuelen 1000 (Ethernet)
    RX packets 765341 bytes 979134471 (933.7 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 490486 bytes 39054386 (37.2 MiB)
    TX errors 0 dropped 58 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 9294 bytes 481659 (470.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 9294 bytes 481659 (470.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

3. [tracert google.com](https://www.google.com/traceroute)

Purpose: Traces the path data takes across multiple routers/hops to reach the destination server, useful for diagnosing where connectivity issues occur.

SCREENSHOT

Terminal output of 'tracert google.com' showing each hop to the destination

```
(mahnoor@kali)-[~/Desktop/dg]
$ traceroute google.com
traceroute to google.com (142.250.203.174), 30 hops max, 60 byte packets
 1  192.168.61.2 (192.168.61.2)  0.353 ms  0.230 ms  0.163 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * * rails_exchange_de
 8  * * * ep.sh
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * * 02608180...
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * * 02608180...
28  * * *
29  * * *
30  * * *
```

Network Simulation – Cisco Packet Tracer

A simple network was built and tested in Cisco Packet Tracer (free tool from Cisco NetAcad) to apply the theoretical concepts learned in Part 1.

Step 1: Installation

Created a free Cisco NetAcad account and downloaded/installed Cisco Packet Tracer.

SCREENSHOT

Packet Tracer installed and open on the desktop



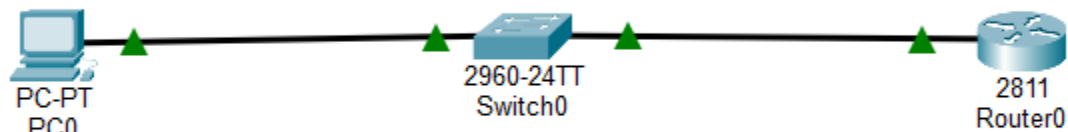
Step 2 & 3: Building and Connecting the Network

A basic topology was built consisting of a PC connected to a Switch, which is connected to a Router, using Copper Straight-Through cables:

- PC → Switch (Copper Straight-Through)
- Switch → Router (Copper Straight-Through)

SCREENSHOT

Packet Tracer workspace showing PC – Switch – Router topology with cable connections



Step 4: IP Configuration

Device	IP Address	Subnet Mask / Gateway
PC	192.168.1.10	Subnet: 255.255.255.0 Gateway: 192.168.1.1
Router	192.168.1.1	Interface configured as default gateway

SCREENSHOT

PC and Router IP configuration windows in Packet Tracer showing the values above

Physical

Config

Desktop

Programming

Attributes

Configuration

X

Interface

FastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.1.10

Subnet Mask

255.255.255.0

Default Gateway

192.168.1.1

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::260:2FFF:FE37:827A

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fastethernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config)#
```

Step 5: Connectivity Test

From the PC's Command Prompt (within Packet Tracer), the command ping 192.168.1.1 was run to confirm that the PC could successfully reach the Router. A successful reply confirms the network was configured correctly.

SCREENSHOT

Packet Tracer PC command prompt showing successful 'ping 192.168.1.1' replies

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

PART 4

Key Learnings

Completing Week 1 of the DG Interns Hub Cybersecurity Internship provided a solid, practical grounding in networking fundamentals. The key outcomes were:

- A clear understanding of how the OSI and TCP/IP models describe the journey of data across a network.
- Familiarity with core protocols (HTTP/HTTPS, FTP, DNS, DHCP) and their real-world roles.
- The ability to interpret IP addressing, distinguish public vs. private IPs, and grasp basic subnetting.
- Hands-on experience running real diagnostic commands (ping, ipconfig/ifconfig, tracert).
- Practical experience designing and testing a simple network topology in Cisco Packet Tracer.

This foundation in networking is essential for the cybersecurity path ahead — every future topic, from firewalls to penetration testing, builds directly on these fundamentals.